

PORTLAND INTL, OR, USA

WMO: 726980

Lat: 45.596N

Lon: 122.609W

Elev: 19

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24229

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	25.9	29.4	11.0	9.6	31.7	16.7	12.8	34.7	30.8	37.5	27.4	39.3	9.5	120	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.6	91.7	67.4	87.5	66.3	83.8	64.9	69.4	87.1	67.8	84.6	66.2	81.3	10.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	85.9	74.8	61.4	81.4	73.2	60.1	77.6	71.6	33.4	87.3	32.0	84.4	30.7	81.3	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.4	19.5	17.2	DB	21.2	99.2	5.2	3.4	17.5	101.7	14.5	103.7	11.6	105.6	7.8	108.1	(4)
(5)			WB	19.5	72.5	5.0	1.8	15.9	73.7	13.0	74.7	10.2	75.7	6.6	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	41.7	43.7	48.0	52.4	59.0	63.9	70.0	70.4	65.1	55.3	47.1	41.3		
	DBStd	11.66	5.74	5.37	4.89	5.38	6.08	5.80	5.31	4.79	5.34	4.98	5.74	5.85		
	HDD50	983	262	183	98	31	3	0	0	0	0	11	119	276		
	HDD65	4179	724	597	528	379	206	88	12	7	64	302	536	736		
	CDD50	2764	4	6	35	104	281	417	620	631	453	176	32	5		
	CDD65	484	0	0	0	2	19	54	167	173	67	2	0	0		
	CDH74	4607	0	0	1	47	273	578	1566	1480	639	23	0	0		
	CDH80	1746	0	0	0	8	79	213	641	591	212	2	0	0		
	WSAvg	7.3	9.0	8.4	7.6	6.9	6.4	6.7	6.9	6.5	5.9	6.2	7.8	9.1		
	PrecAvg	46.3	6.8	4.5	4.9	3.7	2.9	2.0	0.6	0.7	1.7	4.3	7.0	7.4		
(15) Precipitation	PrecMax	73.0	13.2	11.3	8.5	7.5	5.8	4.6	2.3	3.3	5.9	9.5	15.8	15.6		
	PrecMin	35.5	2.4	0.4	1.9	1.5	0.2	0.4	0.0	0.0	0.0	0.9	2.4	2.8		
	PrecStd	8.3	2.5	2.2	1.7	1.5	1.5	0.9	0.5	0.7	1.2	2.1	3.1	3.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.9	60.5	70.2	80.1	87.3	94.0	97.9	96.3	91.3	77.2	64.5	58.5		
		MCWB	54.3	51.0	55.4	60.1	63.2	68.0	70.0	68.3	65.6	59.7	58.2	55.1		
	2%	DB	54.3	57.0	64.6	72.8	81.5	86.0	91.8	90.9	86.0	71.5	60.3	54.2		
		MCWB	50.2	49.6	52.2	57.2	61.6	65.1	68.2	67.1	63.6	57.6	55.1	51.0		
	5%	DB	51.8	54.3	60.1	67.0	75.9	80.4	86.9	86.6	81.5	67.6	57.7	51.3		
		MCWB	48.2	48.0	50.4	53.9	59.8	63.1	67.2	66.7	63.1	56.7	53.4	48.4		
	10%	DB	49.5	51.6	56.6	62.3	70.8	75.6	82.5	82.3	76.8	63.8	55.2	49.0		
		MCWB	46.2	46.3	49.1	52.1	57.8	61.2	65.4	65.3	61.9	56.0	51.4	45.7		
	0.4%	WB	55.2	54.9	57.3	61.7	66.0	69.8	72.7	70.7	68.7	64.1	59.9	55.6		
		MCDB	57.5	57.8	67.2	77.0	82.6	89.0	92.6	90.1	85.0	71.8	62.5	58.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	51.2	51.4	54.1	58.0	63.3	66.4	69.9	68.9	66.4	60.9	56.3	51.5		
		MCDB	53.3	54.8	60.7	70.3	77.3	83.1	87.5	86.7	80.8	67.2	59.2	53.6		
	5%	WB	48.8	49.0	52.0	55.4	61.0	64.0	68.0	67.5	64.6	59.0	54.1	48.9		
		MCDB	51.2	53.1	57.8	64.6	73.3	77.8	84.7	84.0	77.1	64.3	56.8	51.1		
	10%	WB	46.5	47.1	50.3	53.2	58.8	62.1	66.1	65.9	63.0	57.2	52.2	46.3		
		MCDB	49.0	50.9	55.8	60.3	69.2	73.9	81.4	80.5	73.8	61.8	54.7	48.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	12.7	14.9	16.4	17.8	18.5	21.9	21.6	20.2	15.4	10.9	9.0		
		MCDBR	12.0	16.8	22.2	26.5	28.5	28.2	29.6	29.4	29.4	23.1	13.0	10.7		
		MCWBR	8.8	10.2	11.9	12.7	12.4	11.6	11.3	10.7	11.0	11.0	8.3	8.6		
		MCDBR	10.9	13.9	18.1	23.0	24.9	25.5	27.5	26.6	24.8	16.3	10.5	10.3		
	5% WB	MCWBR	8.9	9.4	10.9	11.8	11.6	11.2	11.2	10.4	10.4	9.3	7.8	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.317	0.330	0.362	0.377	0.362	0.350	0.357	0.344	0.334	0.322	0.326		
	taud		2.531	2.552	2.524	2.400	2.375	2.432	2.490	2.486	2.505	2.543	2.540	2.477		
	Ebn at Noon		251	277	287	285	283	287	289	283	279	266	248	234		
	Edh at Noon		21	24	29	35	37	36	33	32	29	24	20	20		
(44) All-Sky Solar Radiation	RadAvg		367	633	921	1342	1708	1888	2150	1836	1372	793	428	299		
	RadStd		60	89	122	126	165	177	151	124	120	99	58	45		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+70
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+47

Nomenclature: See separate page